

LSC Implementation Templates

LSC Archive Structure Plan

Status: Step 3 implementation artifact

Purpose: local/GitHub/Zenodo freeze package preparation plan only

Publishing: not performed

1. Local Archive Structure

```
LSC_FREEZE_PACKAGE/  
  README.md  
  MANIFEST.csv  
  CHECKSUMS.sha256  
  canonical/  
    LSC_CANONICAL_CORE.md  
    LSC_ACTIVE_EQUATION_SET.md  
    LSC_FREEZE_BOUNDARY.md  
  validation/  
    LSC_VALIDATION_PROTOCOL.md  
    LSC_M4_DETECTOR_SYSTEMATICS_BASELINE.md  
    LSC_COVARIANCE_AND_DATA_POLICY.md  
    LSC_EXTERNAL_VETO_CHECKLIST.md  
    LSC_PREREGISTRATION_POLICY.md  
    LSC_FAILURE_MODES_AND_RISK_MATRIX.md  
  implementation/  
    LSC_DATA_REQUIREMENT_MATRIX.md  
    LSC_M0_M6_SPECIFICATION_TEMPLATES.md  
    LSC_M4_IMPLEMENTATION_SCHEMA.md  
    LSC_COVARIANCE_PLACEHOLDER_AND_SENSITIVITY_TEMPLATE.md  
    LSC_LOO_EXECUTION_PROTOCOL.md  
    LSC_CROSS_EXPERIMENT_TRANSFER_PROTOCOL.md  
    LSC_EXTERNAL_VETO_DATA_REQUEST_CHECKLIST.md  
    LSC_BEST2_PREREGISTRATION_FORM.md  
  audits/  
    original/  
    extracted/  
  sources/  
    source_inventory.csv  
    zenodo_metadata/  
  reports/  
    PDFs/  
  provenance/  
    lineage.md  
    deprecated_components.md  
    source_map.csv
```

2. GitHub Release Preparation

Prepare but do not publish:

- freeze tag name proposal;
- release notes;
- changelog;
- checksum manifest;
- source inventory;
- no generated claims beyond frozen documents.

3. Zenodo Freeze Package Preparation

Prepare but do not upload:

- archive zip;
- `zenodo.json`;
- DOI relation notes;
- concept DOI continuity statement;
- included files list;
- excluded files list.

4. Reproducibility Structure

Required:

- all Markdown source files;
- all generated PDFs;
- source inventory;
- audit inventory;
- checksum manifest;
- scripts used for generation, if any;
- explicit note that placeholders are non-validation-grade.

5. Provenance Structure

Must preserve:

- Step 1 canonicalization decisions;
- Step 2 validation rules;
- Step 3 implementation templates;
- original audits;
- extracted audit text;
- Zenodo metadata;
- source repository references.

6. Audit Traceability

Every archive must answer:

- what became canonical;
- what became deprecated;
- what validation is required;
- what data are missing;
- what claims are prohibited;
- what negative outcomes mean.

LSC BEST-2 Preregistration Form

Status: Step 3 implementation artifact

Purpose: preregistration-ready freeze form

1. Project Identity

- project: LSC Pre-BEST-2 Freeze Validation
- theory status: unvalidated phenomenological framework
- active freeze state: LSC 6.3.0
- no LSC 6.4 creation: confirmed

2. Frozen Equations

Reference:

- Step 1 LSC_ACTIVE_EQUATION_SET.md

Checklist:

- ☐ active equation set copied verbatim;
- ☐ deprecated equation families excluded;
- ☐ dimensional-review items labeled;
- ☐ no new equation introduced.

3. Frozen Parameter Structure

Checklist:

- ☐ all active parameters listed;
- ☐ nuisance parameters listed;
- ☐ bounds specified;
- ☐ tensor orientation specified if anisotropy tested;
- ☐ `f0_a` and `fA_a` functional forms frozen if used.

4. Frozen Datasets

Checklist:

- ☐ BEST data version listed;
- ☐ GALLEX data version listed;

- ☐ SAGE data version listed;
- ☐ BEST-2 target data access conditions listed;
- ☐ external veto data versions listed;
- ☐ missing data declared.

5. Frozen Validation Hierarchy

M0-M6 must all be included:

- ☐ M0;
- ☐ M1;
- ☐ M2;
- ☐ M3;
- ☐ M4;
- ☐ M5;
- ☐ M6.

6. Prohibited Post-Hoc Actions

Prohibited:

- adding mechanisms;
- adding equations;
- adding tensor structures;
- changing tensor orientation after data inspection;
- suppressing covariance limitations;
- dropping M4;
- dropping M3;
- changing success/failure criteria;
- using AI consensus as evidence.

7. Allowed Statistical Interpretations

Allowed:

- exploratory sensitivity;
- validation-grade comparison only if covariance and data requirements are met;
- unresolved ambiguity;
- failure;
- negative result.

Not allowed:

- discovery claim;
- sterile-neutrino exclusion claim;
- confirmed new physics claim.

8. Failure Conditions

Failure or major weakening if:

- M4 reproduces M6;
- M1/M2 perform as well as M5/M6 after penalties;
- LOO unstable;
- cross-experiment transfer fails;
- covariance unavailable for claimed validation;
- external veto excludes required parameter space.

9. Publication Discipline

Every release must include:

- data versions;
- covariance status;
- M0-M6 results;
- M4 status;
- LOO results;
- external veto status;
- negative outcomes;
- unresolved blockers.

LSC Covariance Placeholder And Sensitivity Template

Status: Step 3 implementation artifact

1. Core Label

All placeholder covariance outputs must be labeled:

NON-VALIDATION-GRADE

2. Placeholder Policy

Allowed placeholders:

- diagonal covariance from published summary uncertainties;
- block-diagonal covariance by experiment;
- conservative correlated-error envelope;
- source-normalization correlation envelope;
- cross-section shared uncertainty envelope.

Forbidden:

- treating placeholders as final covariance;

- using placeholders for discovery, exclusion or strong preference claims;
- suppressing missing-covariance caveats.

3. Machine-Readable Placeholder Template

```

covariance_policy:
  validation_grade: NON-VALIDATION-GRADE
  covariance_type: diagonal | block_diagonal | conservative_envelope
  source:
    - published_summary_uncertainties
  missing:
    - full_systematic_covariance
    - extraction_level_correlations
  allowed_claims:
    - exploratory_sensitivity
  prohibited_claims:
    - likelihood_grade_validation
    - discovery
    - exclusion

```

4. Sensitivity Analysis Structure

Every exploratory run must compare:

- diagonal errors;
- high-correlation envelope;
- low-correlation envelope;
- source-normalization shared shift;
- cross-section shared shift;
- zone-correlated extraction shift.

5. Missing-Data Propagation

Every result table must include:

- missing covariance fields;
- affected model comparisons;
- blocked claims;
- sensitivity to placeholder choice;
- validation grade.

6. Statistical Caveat Policy

Required statement for placeholder covariance:

This result is NON-VALIDATION-GRADE because full statistical and systematic covariance is un

LSC Cross-Experiment Transfer Protocol

Status: Step 3 implementation artifact

1. Purpose

Cross-experiment transfer tests whether a frozen model generalizes beyond one experiment.

2. Transfer Directions

Fit Source	Prediction Target
BEST	GALLEX
BEST	SAGE
GALLEX/SAGE	BEST
51Cr sources	37Ar source
BEST/GALLEX/SAGE	future BEST-2
BEST inner	BEST outer
BEST outer	BEST inner

3. Transfer Conditions

Transfer is valid only if:

- mapping is frozen before fitting;
- geometry differences are documented;
- isotope/source differences are documented;
- covariance limitations are stated;
- M0-M6 are evaluated under the same transfer;
- no target-specific retuning is allowed.

4. Transfer Failure

Transfer fails if:

- model fits source experiment but misses target experiment;
- parameters require unrelated values per experiment;
- M4 transfers as well or better than M6;
- isotope transfer fails without explanation;
- covariance sensitivity changes conclusion.

5. Isotope Consistency Checks

Required:

- 51Cr-trained behavior tested against 37Ar where possible;

- source-line and cross-section uncertainties propagated;
- isotope-specific systematics tested under M2 and M4.

6. Geometry Consistency Checks

Required:

- source position mapping;
- detector zone mapping;
- geometry uncertainty propagation;
- separation of source geometry from detector artifact.

7. External Veto Dependencies

Transfer success is not sufficient if:

- KATRIN veto fails;
- IceCube/SME tensor constraint fails;
- solar or standard oscillation consistency fails;
- detector calibration literature supports a simpler explanation.

LSC Data Requirement Matrix

Status: Step 3 implementation artifact

Scope: dataset infrastructure preparation only

No theory extension: yes

1. Classification Key

Class	Meaning
ESSENTIAL	Required for likelihood-grade validation, core falsification, or required veto
IMPORTANT	Needed for robust interpretation but not always a hard blocker
OPTIONAL	Useful for completeness, plotting, or secondary review
CURRENTLY UNAVAILABLE	Not available in the current local/public package

2. Machine-Readable Matrix

Resource	Availability	Machine-readable status	Covariates	Extraction-level availability	Orientation availability	Exposure-level timestamp availability	Relevance	Classification
BEST	summary values present in local package	partial JSON/CSV summary	unavailable	unavailable	unavailable	unavailable	primary two-zone Galilium validation target	ESSENTIAL
GALLEX	summary values present	partial JSON/CSV summary	unavailable	unavailable	unavailable	incomplete	cross-experiment transfer	IMPORTANT
SAGE	summary values present	partial JSON/CSV summary	unavailable	unavailable	unavailable	incomplete	cross-experiment and isotope transfer	IMPORTANT
BEST-2	future/planned	unavailable	unavailable	unavailable	unavailable	unavailable	future decisive pre-registered validation	ESSENTIAL / CURRENTLY UNAVAILABLE
KATRIN	public literature exists	not integrated locally as constraints	not integrated	not applicable	not applicable	not applicable	external energy-response veto	ESSENTIAL

Resource	Availability	Machine-readable status	Machine-readable availability	Covariance level	Extraction-level availability	Orientation-level availability	Exposure-level timestamp	Relevance	Classification
IceCube	public constraint literature exists	not integrated locally as tensor bounds	not integrated	not applicable	not applicable	not applicable	not applicable	anisotropy / SME-style veto	ESSENTIAL if tensor is physical/celestial
solar-neutrino constraints	public literature exists	not integrated locally	not integrated	not applicable	not applicable	not applicable	not applicable	consistent veto for universal response	IMPORTANT
SME/Lorentz constraint literature	public literature exists	not integrated locally	not integrated	not applicable	not applicable	not applicable	not applicable	tensor-background veto	ESSENTIAL if tensor interpretation exceeds detector-local response
detector calibration/branching/systematics literature	partially referenced/systematics	not consolidated	unavailable	unavailable	unavailable	unavailable	unavailable	M4 baseline construction	ESSENTIAL

3. Required Future Data Fields

BEST / BEST-2

- source activity calibration;
- source position and source extension;

- zone geometry;
- extraction-by-extraction table;
- counter efficiencies;
- background model;
- statistical covariance;
- systematic covariance;
- detector orientation;
- exposure timestamps;
- environmental logs;
- machine-readable baseline predictions.

GALLEX

- source run metadata;
- exact source coordinates;
- exposure-level rates;
- detector response/covariance;
- uncertainty convention;
- source activity details.

SAGE

- run-by-run source exposure;
- isotope-specific source metadata;
- counter-level likelihood inputs;
- ^{37}Ar impurity table;
- covariance/systematics table.

External Vetoes

- KATRIN energy-response constraints;
- IceCube anisotropy/Lorentz/SME bounds;
- solar-neutrino consistency tables;
- standard oscillation fit constraints;
- calibration/cross-section uncertainty literature.

4. Implementation Rule

Until ESSENTIAL missing data are available, outputs must be labeled:

NON-VALIDATION-GRADE / EXPLORATORY ONLY

LSC External Veto Data Request Checklist

Status: Step 3 implementation artifact

1. KATRIN

Required data:

- energy-response constraints;
- beta-spectrum systematic model;
- machine-readable uncertainty information where public;
- literature bounds relevant to energy-scale distortion.

Why it matters:

- constrains universal energy-response interpretations.

Constrained structure:

- trace response / energy-response transfer.

Public access:

- literature public; machine-readable integration not present locally.

2. IceCube

Required data:

- anisotropy constraints;
- Lorentz-violation or SME-related neutrino bounds;
- systematic treatment of detector anisotropy;
- machine-readable limits where public.

Why it matters:

- constrains physical/celestial tensor interpretations.

Constrained structure:

- traceless anisotropy if interpreted beyond detector-local response.

Public access:

- literature public; mapped dataset not integrated locally.

3. SME / Lorentz Literature

Required data:

- published coefficient bounds;
- mapping rules for any fixed tensor interpretation;
- energy and direction dependence of limits.

Why it matters:

- prevents hidden Lorentz-violation conflict.

Constrained structure:

- tensor-background interpretations.

Public access:

- public literature; mapping not implemented.

4. Solar-Neutrino Fits

Required data:

- solar Gallium consistency constraints;
- global solar-neutrino fit summaries;
- uncertainty treatment.

Why it matters:

- tests whether response is source-experiment-specific or universal.

Constrained structure:

- scalar/trace response.

Public access:

- literature public; machine-readable integration absent.

5. Standard Oscillation Constraints

Required data:

- three-flavor global fit constraints;
- sterile-neutrino benchmark literature;
- L/E and geometry inputs for source experiments.

Why it matters:

- keeps LSC within known oscillation constraints.

Constrained structure:

- any survival/transition behavior.

Public access:

- literature public; local integration incomplete.

6. Detector Calibration/Systematics Literature

Required data:

- gallium cross-section uncertainty;
- source activity calibration uncertainty;
- extraction efficiency literature;
- counter efficiency uncertainty;

- background modeling literature.

Why it matters:

- powers M2 and M4.

Constrained structure:

- trace response and detector/systematics mimicry.

Public access:

- partially public; not consolidated into machine-readable M4 package.

LSC Leave-One-Out Execution Protocol

Status: Step 3 implementation artifact

1. Purpose

Leave-one-out testing prevents anchor-fit overclaiming and tests parameter stability.

2. Required Inputs

- dataset table;
- model specification M0-M6;
- covariance or placeholder covariance label;
- parameter bounds;
- nuisance parameter policy;
- penalty method.

3. Required Omission Runs

Omission	Prediction Target
omit BEST inner	BEST inner
omit BEST outer	BEST outer
omit BEST ratio	BEST ratio
omit GALLEX combined	GALLEX combined
omit GALLEX Cr1	GALLEX Cr1
omit GALLEX Cr2	GALLEX Cr2
omit SAGE 51Cr	SAGE 51Cr
omit SAGE 37Ar	SAGE 37Ar
omit BEST group	BEST group
omit GALLEX group	GALLEX group
omit SAGE group	SAGE group

4. Execution Steps

1. Load frozen model template.
2. Load dataset and covariance status.
3. Select one omission target.
4. Fit or evaluate on remaining data only.
5. Predict omitted target.
6. Record residual and uncertainty.
7. Record parameter changes.
8. Repeat for M0-M6.
9. Compare penalty-adjusted performance.
10. Classify stability.

5. Stability Classification

Class	Meaning
stable	predictions remain within preregistered tolerance and parameters remain bounded
weakly stable	predictions degrade but model remains interpretable
unstable	parameters swing or predictions fail significantly
non-identifiable	fit cannot distinguish model components
non-validation-grade	covariance/data limitations block validation claim

6. Failure Handling

If LOO fails:

- do not claim validation;
- report instability;
- compare whether M4/M1/M2 are more stable;
- reduce model complexity only in a future preregistered version;
- do not add post-hoc mechanisms.

LSC M0-M6 Specification Templates

Status: Step 3 implementation artifact

Purpose: formal model-template definitions for future validation execution

Shared Rules

All models must report:

- datasets used;
- parameters and nuisance parameters;
- allowed freedoms;
- forbidden freedoms;
- penalty method;
- covariance status;
- comparison targets;
- allowed claims;
- prohibited claims.

No template may introduce new theory.

M0 - Null / Published Baseline

Field	Specification
purpose	published or standard baseline expectation
parameters	none beyond published baseline inputs
allowed freedoms	documented experimental uncertainties only
forbidden freedoms	hidden normalization tuning
required penalties	none or fixed-baseline accounting
required datasets	BEST/GALLEX/SAGE baseline expectations
comparisons	all models vs M0
allowed claims	residual exists relative to baseline
prohibited claims	LSC support by residual alone

M1 - Scalar Normalization Only

Field	Specification
purpose	test global scale / normalization explanation
parameters	scalar normalization nuisance
allowed freedoms	one preregistered scale parameter or documented source-specific scale
forbidden freedoms	zone-specific tensor-like behavior
required penalties	scalar parameter penalty
required datasets	rate ratios and covariance
comparisons	M1 vs M5/M6
allowed claims	trace may be normalization-like
prohibited claims	directional structure

M2 - Cross-Section / Source-Systematics Only

Field	Specification
purpose	test nuclear/source explanation
parameters	source activity, cross-section and isotope nuisance terms
allowed freedoms	literature-supported uncertainties
forbidden freedoms	arbitrary post-hoc source shifts
required penalties	nuisance and uncertainty penalties
required datasets	source lines, source activity, cross-section uncertainties
comparisons	M2 vs M5/M6
allowed claims	source/cross-section can explain trace if supported
prohibited claims	LSC preference without outperforming M2

M3 - Sterile-Neutrino Benchmark

Field	Specification
purpose	required physics competitor
parameters	sterile benchmark parameters as preregistered
allowed freedoms	standard benchmark scan or fixed literature benchmark
forbidden freedoms	omitting M3 from final comparison
required penalties	complexity/scan penalty
required datasets	L/E mapping, geometry, rates, covariance
comparisons	M3 vs M5/M6
allowed claims	LSC compares to sterile baseline
prohibited claims	sterile excluded unless full analysis supports it

M4 - Detector-Systematics Benchmark

Field	Specification
purpose	required mundane competitor
parameters	calibration, source, extraction, geometry, background, environment, covariance and orientation artifacts

Field	Specification
allowed freedoms	physically motivated detector/systematics terms
forbidden freedoms	treating M4 as optional
required penalties	nuisance and complexity penalties
required datasets	calibration, extraction, background, geometry, covariance metadata
comparisons	M4 vs M5/M6
allowed claims	detector/systematics can mimic LSC if it fits
prohibited claims	LSC preference before M4 is evaluated

M5 - LSC Trace-Only

Field	Specification
purpose	test scalar trace response
parameters	frozen trace response amplitude and nuisance terms
allowed freedoms	preregistered scalar template only
forbidden freedoms	directional/tensor terms
required penalties	trace parameter penalty
required datasets	rate ratios, covariance, baseline expectations
comparisons	M5 vs M1/M2/M4/M6
allowed claims	trace template improves fit if validated
prohibited claims	anisotropy or sidereal signal

M6 - LSC Trace + Traceless Anisotropy

Field	Specification
purpose	full frozen LSC detector-response test
parameters	trace amplitude, anisotropic amplitude, frozen tensor/orientation structure, nuisance terms
allowed freedoms	only preregistered trace/traceless template
forbidden freedoms	post-hoc tensor orientation or added terms
required penalties	full complexity penalty
required datasets	covariance, orientation/timestamps, geometry, exposure data
comparisons	M6 vs M0-M5

Field	Specification
allowed claims	candidate support only if all validation criteria pass
prohibited claims	discovery, sterile exclusion, or validation without M4/covariance/vetoos

LSC M4 Implementation Schema

Status: Step 3 implementation artifact

Purpose: detector/systematics competitor schema

1. Core Rule

M4 is the full detector/systematics competitor. It must be implemented before M6 preference can be claimed.

2. Machine-Readable Top-Level Structure

```

model_id: M4
model_name: detector_systematics_baseline
validation_grade: NON-VALIDATION-GRADE until covariance and metadata are complete
components:
  calibration_drift: {}
  source_normalization_uncertainty: {}
  detector_geometry_asymmetry: {}
  extraction_efficiency_asymmetry: {}
  environmental_background_drift: {}
  covariance_incompleteness: {}
  timing_artifacts: {}
  orientation_artifacts: {}
  background_subtraction_uncertainty: {}
outputs:
  predicted_ratios: []
  nuisance_parameters: []
  covariance_assumptions: []
  comparison_scores: []

```

This is an implementation schema, not a new physics model.

3. Component Schemas

Calibration Drift

```
calibration_drift:
  status: placeholder
  inputs_required:
    - calibration_run_id
    - timestamp
    - calibration_value
    - uncertainty
  sensitivity_hooks:
    - drift_amplitude_scan
    - time_split_test
```

Source Normalization Uncertainty

```
source_normalization_uncertainty:
  status: placeholder
  inputs_required:
    - source_activity
    - source_activity_uncertainty
    - calibration_method
    - isotope
  sensitivity_hooks:
    - source_scale_scan
    - isotope_split_test
```

Detector Geometry Asymmetry

```
detector_geometry_asymmetry:
  status: placeholder
  inputs_required:
    - detector_geometry_file
    - source_position
    - zone_boundaries
    - dead_volume_map
  sensitivity_hooks:
    - geometry_perturbation_scan
    - source_position_shift_test
```

Extraction Efficiency Asymmetry

```
extraction_efficiency_asymmetry:
  status: placeholder
  inputs_required:
    - extraction_id
```

```

- zone_id
- efficiency
- efficiency_uncertainty
sensitivity_hooks:
- zone_efficiency_scan
- run_by_run_efficiency_test

```

Environmental Background Drift

```

environmental_background_drift:
  status: placeholder
  inputs_required:
    - timestamp
    - temperature
    - pressure
    - radon_level
    - electronics_state
  sensitivity_hooks:
    - environmental_regression
    - day_night_split
    - seasonal_split

```

Covariance Incompleteness

```

covariance_incompleteness:
  status: placeholder
  inputs_required:
    - covariance_matrix
    - covariance_source
    - missing_correlation_policy
  sensitivity_hooks:
    - diagonal_vs_correlated_comparison
    - conservative_correlation_envelope

```

Timing Artifacts

```

timing_artifacts:
  status: placeholder
  inputs_required:
    - exposure_start
    - exposure_end
    - extraction_time
    - counting_time
  sensitivity_hooks:
    - time_scramble
    - exposure_phase_split

```

Orientation Artifacts

```
orientation_artifacts:
  status: placeholder
  inputs_required:
    - detector_azimuth
    - detector_tilt
    - latitude
    - longitude
    - source_orientation
  sensitivity_hooks:
    - lab_frame_vs_celestial_frame_test
    - orientation_scramble
```

Background Subtraction Uncertainty

```
background_subtraction_uncertainty:
  status: placeholder
  inputs_required:
    - background_rate
    - background_uncertainty
    - background_model
    - background_run_id
  sensitivity_hooks:
    - background_scale_scan
    - background_only_validation
```

4. Required M4 Report Fields

Each M4 execution must output:

- included components;
- missing components;
- nuisance parameter count;
- covariance assumption;
- fit score;
- penalty-adjusted score;
- comparison vs M5/M6;
- whether M4 reproduces M6-level structures.